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Intelligent framework for coagulant dosing optimization in an industrial-scale seawater reverse osmosis desalination plant

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ABSTRACT

Efficient and reliable desalination through seawater reverse osmosis (SWRO) mandates optimized pre-treatment strategies to minimize organic and inorganic fouling. Coagulation, the process of agglomerating colloidal particles using chemical coagulants, in combination with media filtration to reduce colloidal fouling on reverse osmosis membranes is commonly used in seawater pretreatment. Due to its inherent complexity and the absence of physical models to quantify the efficiency of coagulation, overdosing of coagulants is ubiquitously observed to maintain filtered water quality. To address this problem, we use Artificial neural networks (ANNs) to optimize coagulant dosing by predicting the SDI after chemical dosing. The model is developed by using large-scale plant data comprising of different seawater physical parameters and plant operational data including pH, SDI, turbidity, coagulant dosing rate, and flocculant dosing rate. By using feature engineering, selection, and our domain knowledge, new input parameters are derived, irrelevant parameters are eliminated, and these are used as inputs to train the model. The developed ANNs model achieved a prediction accuracy of 95% also outperforms other machine learning methods, and upon industrialization it reduced annual coagulant consumption by 11.7% when implemented in a commercial SWRO plant producing 216,000 m^3/day of desalinated water.

1. Introduction

Freshwater scarcity, one of the biggest challenges of our time, continues to expand across different regions (Elimelech & Phillip, 2011; Ghaffour, Missimer, & Amy, 2013b). This issue is escalated further by the continuous growth of population in inherently water stressed regions (Elimelech & Phillip, 2011; Mekonnen & Hoekstra, 2016). To address this, seawater reverse osmosis (SWRO), the most efficient technology to generate freshwater from seawater (Altmann et al., 2022; Qasim, Badrelzaman, Darwish, Darwish, & Hilal, 2019), is relied upon in many regions including the Middle East, Africa, the US west coast, and the Southern part of Europe (Ghaffour, Missimer, & Amy, 2013a; Gude, Nirmalakhandan, & Deng, 2011). The technology is based on the separation of water and inorganic salt solute from seawater through a semi-permeable membrane (Das, Arunachalam, Ahmad, Manalastas, & Mishra, 2019; Gonzalez-Gil et al., 2021). SWRO utilizes energy in form of potential energy to counter the osmotic gradient of seawater to produce low salinity permeate and concentrated brine. To further increase the efficiency and reliability of the SWRO process, its operation is complemented with rigorous pre-treatment technologies and processes from bar screen, media filtration, dissolved air flotation, disinfection, coagulation, de-chlorination, and acidification, among others (Altmann & Das, 2021; Voutchkov, 2010, 2012).

The SWRO process has undergone several enhancements over the last decade, it still remains an energy and chemical-intensive process and this intensifies environmental stress (Elimelech & Phillip, 2011; Ghaffour et al., 2013b; Hasanin et al., 2023; Karabelas, Koutsou, Kostoglou, & Sioutopoulos, 2018; Voutchkov, 2018). The pressing demand to squeeze energy consumption to its near theoretical level has prompted research primarily focusing on improving the energy efficiency of the process. The use of novel materials to develop next generation desalination membranes with ultra-high water permeability and high water–solute selectivity has been thoroughly explored (Werber, Osuji, & Elimelech, 2016). Despite substantial research efforts towards the development of highly efficient membranes, the resulting reduction in energy consumption has been minimal; this calls for a paradigm shift in the current desalination research approach towards rational system design, as well as the optimization of process operations (Patel et al., 2020; Yousry, Ridwan, Altmann, Rousseva, Azab, & Das, 2022). While many physical models could be utilized to achieve this, however, the complexity of the processes engender inefficient use of computation usage and inaccurate results (Dudchenko & Mauter, 2020).

One of the sub-processes that is phenomenologically complex, has significant impact on the efficiency of the SWRO process and requires

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optimization is coagulation of seawater during pretreatment. The coagulation process is employed to remove suspended solids by destabilizing the particle using chemicals to agglomerate the particles (Gonzalez-Gil et al., 2021). The coagulant demand of seawater depends on certain parameters such as its salinity, pH, presence of humic substances and algal organic matter (Duan, Niu, Shi, Wilson, & Graham, 2009; Shutova et al., 2016). For an effective pretreatment, coagulant dosing needs optimization. Dosing lesser than what is required will result in ineffective removal of organic matter, resulting in biofouling on the membrane (Tabatabai, Schippers, & Kennedy, 2014), while dosing excess will result in inorganic fouling on the RO membrane which can lead to membrane oxidation damage (Chesters et al., 2013; Gabelich, Yun, & Coffey, 2002; Gonzalez-Gil et al., 2021). The coagulant dosing should be closely linked to the presence of residual Fe^{3+} in the filtered water and its turbidity. While physical models for two well-defined charged particles in an aqueous environment are ubiquitous (Israelachvili, 2011; Smith, Borkovec, & Trefalt, 2020), the physical model of this specific process is inherently convoluted due to the myriad interactions between suspended solids in an ever fluctuating aqueous environment (Dishon, Zohar, & Sivan, 2009; Israelachvili, 2011; Misra, de Souza, Blankschtein, & Bazant, 2019; Pillai, Santana, Das, Shrestha, Manalastas, & Mishra, 2020; Ridwan, Shrestha, Maharjan, & Mishra, 2022; Smith et al., 2020; Smith, Maroni, Trefalt, & Borkovec, 2019; Trefalt, Behrens, & Borkovec, 2016). Hence, a simple yet accurate technique to optimize this is a rarity and requires meticulous development.

In recent years, a number of studies have utilized algorithms to optimize the coagulation process including the use of fuzzy dynamic modeling (Bello, Hamam, & Djouani, 2014), artificial neural networks (ANNs) (Al-Abri, Al Anezi, Dakheel, & Hilal, 2010; Kennedy, Gandomi, & Miller, 2015), and hybrid methods (Zangooei, Delnavaz, & Asadollahfardi, 2016a; Zangooei, Delnavaz, Asadollahfardi, & Technology, 2016b). However, to our knowledge, there has not been any large-scale demonstrations of such algorithms and further studies on economic impacts of such optimizations. In light of the aforementioned, in the present study, we utilize an artificial neural network-based algorithm to optimize the chemical consumption of coagulants. The developed model is used to predict the Silt Density Index (SDI) of filtered seawater based on the coagulant dosing rate. SDI is a common measurement of the fouling potential of particles in reverse osmosis feedwater (Wei, Laborie, Aim, & Amy, 2012), and is a key indicator of the efficiency of the filtration and coagulation processes. In the beginning, the model is developed and trained using plant data from one of the biggest SWRO desalination plants in the Middle East. Before the development, the specific seawater parameters as inputs were modified by feature engineering and selected based on the feature selection technique and our domain knowledge. We then conduct a parametric optimization for artificial neural network architecture to achieve the highest accuracy between the predicted and test data sets. The model then is deployed and utilized in day-to-day operation in the respective plant. Further, the economic impact of the coagulant optimization is calculated by quantifying the operational cost saving for one-year operation.

2. Literature review

Machine learning (ML) algorithms to optimize coagulant dosing exists, many algorithms used gravitate towards the utilization of artificial neural networks (ANNs). ANNs with different architectures and backpropagation algorithms are used including the inverse neural network (Robenson, Shukor, & Aziz, 2009), Levenberg–Marquardt neural network (Wu & Lo, 2008), generalized regression neural network (Heddam, Bermad, & Dechemi, 2011), radial basis function (Wang, Wu, Deng, Li, & Wang, 2021), adaptive neuro-fuzzy inference system (Narges, Ghorban, Hassan, & Mohammad, 2021; Pandilov & Stojkov, 2019), among others (Achite et al., 2022; Heddam, Dechemi, & treatment, 2015; Jayaweera & Aziz, 2018; Zangooei, Delnavaz, & Asadollahfardi, 2016a; Zangooei, Delnavaz, Asadollahfardi, & Technology, 2016b; Zhang et al., 2019). In addition to ANNs, several other ML methods are used including the linear regression method (Hernandez & Le Lann, 2006), fuzzy linear and non-linear regression methods (Zangooei, Delnavaz, & Asadollahfardi, 2016a; Zangooei, Delnavaz, Asadollahfardi, & Technology, 2016b), k-nearest neighbors (Zhang et al., 2013), and among others (Kim & Parnichkun, 2017b; Wang et al., 2021). Hence, in this section we review several different input parameters' selections, output parameter(s) selection, and ML algorithms that have been used to minimize coagulant dosing in desalination plants.

2.1. Input and output parameter selection

Seawater can be described by several physical properties. Relationship between coagulation process, coagulant chemical, and seawater parameters are rather well known. In general, many studies utilize turbidity, temperature, conductivity, and pH to correlate to coagulation (Taloba, 2022), even electrical conductivity of untreated water are used (Heddam, 2021). The models were developed using several water quality variables selected as input of the models, namely, turbidity, pH, dissolved oxygen, electrical conductivity, and the water temperature. In addition to that, Kim et al. utilize turbidity, temperature, pH, conductivity, historical coagulant dosage to estimate the coagulant dosage (Kim & Parnichkun, 2017a). Zhang et al. used pH, turbidity, and temperature (Zhang et al., 2013). Both resulted in optimization of coagulant dosing rate. In Wu & Lo's work, they utilize the coagulant dosage of the two preceding days, temperature, turbidity, color, pH, coagulation/flocculation, sedimentation and treated water to predict present coagulant dosing (Wu & Lo, 2008). It is interesting to note that these studies rarely employ SDI for predicting or coagulant dosage reduction criteria. In our operational experience, SDI is the most reliable measurement of filtrate quality and criterial for optimum coagulant dosing and widely used by off-taker and membrane supplier for filtrate water specifications. In addition, these methods utilize Pearson correlation coefficient to select the appropriate input parameters. To this end, we utilize Pearson correlation and using wider range of parameters of respective time and historical data to forecast SDI to minimize coagulant consumption.

2.2. Machine learning algorithms

To begin, extremely randomized tree (ERT) and random forest (RF) models are compared with the multiple linear regression (MLR) (Heddam, 2021). The result indicates that ERT performs better than RF and MLR. In addition, utilization of dynamic control algorithm enables a better control to the coagulant dosing, thus minimizing the coagulant consumption without destabilizing the filtration process (Futselaar, Blankert, Spanjer, & Spenkelink, 2013). Additionally, Fang et al. used long–short term memory (LSTM) model, that effectively solved the problem of long-term dependencies of recurrent neural network (Fang, Zhai, Xiong, Zhang, & Gao, 2022). This study demonstrates the efficacy of LSTM model over recurrent neural networks (RNNs). Moreover, Kim et al. tests three different algorithms, namely Multilayer Perceptron (MLP), Adaptive Neuro Fuzzy inference system (ANFIS), and General Regression Neural Network (GRNN) and compared them using correlation coefficient (r) (Kim & Parnichkun, 2017b). They found out that MLP provides better performance on the test. Zhang et al. used Support Vector Regression (SVR) algorithm and coefficient determination to quantify its performance (Zhang et al., 2013). In addition, based on the work by Taloba the ANNs algorithm is used to improve efficiency of various process such as coagulation/flocculation and sedimentation process (Taloba, 2022). Lastly, Wu & Lo used two models, ANFIS and ANNs, and indicates that ANFIS perform better compared to ANNs based on its coefficient of determination (r^2) (Wu & Lo, 2008).

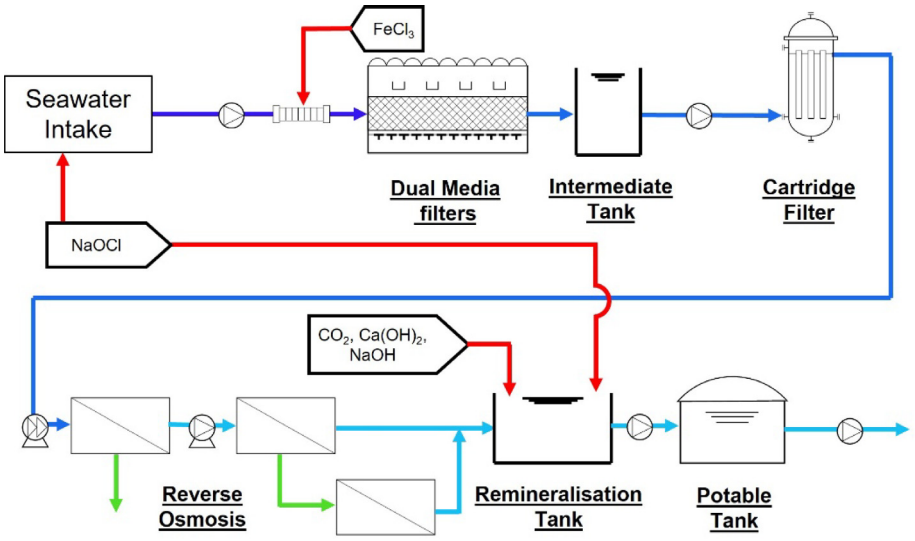


Fig. 1. Schematics of desalination plant.

Although there are a variety of models, input features, and criteria for reducing coagulant dosage, to our knowledge, there is no information on their broad use in an industrial scale desalination facility and the ensuing financial implications. To this end, in the present study we cover the model development, deployment, and its financial implications. We utilized ANNs and optimize its architecture to generate the best model accuracy. The details of the methods are explained in the subsequent section below. It is possible to minimize coagulant dosing more effectively by using these combinations with SDI as the deciding factor.

3. Methods

3.1. Desalination plant description

The desalination plant with production capacity of 216,000 m^3/day is located in the southern part of Saudi Arabia. The desalination plant delivers potable water as product, thereby equipped with potabilization facilities (Fig. 1). In the intake section, the seawater is pumped by the intake pump and intermittently disinfected by dosing Sodium Hypochlorite (NaOCl). Subsequently, the seawater is treated with coagulant and flocculant during the coagulation process before entering the dual media filter (DMF). The seawater then is filtered using a micron-cartridge filter to remove the micron-scale particle. To meet contractual obligations the water reaching the reverse osmosis membrane should have a SDI of less than four. Before passing the seawater to the reverse osmosis section, the seawater is acidified and de-chlorinated. The second pass of the desalination plant starts with sodium hydroxide dosing to maximize boron removal. In the post-treatment process, the permeate water is treated with limewater and CO_2 for remineralization and sodium hydroxide for pH correction.

3.2. Data description

The raw dataset used in this model development comprises of several seawater variables including turbidity, pH, SDI before and after cartridge filter, and coagulant ($FeCl_3$) and flocculant (polymer) dosing rates (see Table 1). The algorithm was trained using three years operational data from the SWRO desalination plant, consisting of 9261 values per parameter. Initially, the data was checked, cleaned, formatted, and validated (including removal of wrong/malformed date format and invalid data such as missing data points). We consider SDI to be valid if the value is above 0, and the polymer dosing rate below 1 ppm. Further, to better understand the relation between SDI and other parameters, we derived additional parameters based on the data. We add additional parameters based on the initial dataset as written below:

(i) Shifting the next value to the current value ($[SDI]_{next,i} = [SDI]_{i+1}$, $[\mathrm{FeCl_3}]_{next,i} = [\mathrm{FeCl_3}]_{i+1}$, $[Polymer]_{next,i} = [Polymer]_{i+1}$)

(ii) Shifting the previous value to the current value ($[SDI]_{prev,i} = [SDI]_{i-1}$, $[\mathrm{FeCl_3}]_{prev,i} = [\mathrm{FeCl_3}]_{i-1}$, $[Polymer]_{prev,i} = [Polymer]_{i-1}$, $[Tur]_{prev,i} = [Tur]_{i-1}$, $[pH]_{prev,i} = [pH]_{i-1}$)

(iii) The differential of the next value ($\Delta[SDI]_{next,i} = [SDI]_{next,i} - [SDI]_i$, $\Delta[\mathrm{FeCl_3}]_{next,i} = [\mathrm{FeCl_3}]_{next,i} - [\mathrm{FeCl_3}]_i$)

(iv) The differential of the previous value ($\Delta[SDI]_{prev,i} = [SDI]_i - [SDI]_{prev,i}$, $\Delta[\mathrm{FeCl_3}]_{prev,i} = [\mathrm{FeCl_3}]_i - [\mathrm{FeCl_3}]_{prev,i}$, $\Delta[Polymer]_{prev,i} = [Polymer]_i - [Polymer]_{prev,i}$, $\Delta[Tur]_{prev,i} = [Tur]_i - [Tur]_{prev,i}$, $\Delta[pH]_{prev,i} = [pH]_i - [pH]_{prev,i}$)

We then perform feature selection to select the best subset of input parameters that correlates better to the output variable. This step, feature selection, is necessary to reduce computation time, improve learning accuracy, avoid redundancy, and provide a better understanding of data and model by removing irrelevant parameters which can lead to poor generalization (Domashova, Emtseva, Fail, & Gridin, 2021); (Orr & Müller, 1998). There are many methods for feature selection, and the widely known are filter, wrapper, and correlation-based methods (Chandrashekar & Sahin, 2014). The filter method is ranking the parameters relative to their significance to the predictor, while the wrapper method wraps the predictor on the search algorithm and provides a group of parameters that gives the highest accuracy. Here, we utilize the correlation-based method–pairwise Pearson correlation (Hall, 1999; Pearson, 1909). This correlation measures the linear relationship between two variables. The Pearson coefficient ranges from −1 to +1, where 0 value indicates no correlation between two variables. Both −1 and +1 are indicating that the two variables correlate perfectly linear with each other (Benesty, Chen, Huang, & Cohen, 2009).

Combining our domain knowledge and evaluation of the pairwise Pearson correlation of those parameters, we selected specific parameters for SDI prediction (Fig. 2). Based on the data and additional calculated parameters, we determine 7 parameters as inputs. The inputs are previous SDI, previous seawater pH, actual seawater pH, previous turbidity, actual turbidity, previous ferric chloride dose, and proposed ferric chloride dose. After selecting and normalizing the subset of input parameters, we split the data for training and testing. We allocate 90% of the data for training and 10% for testing.

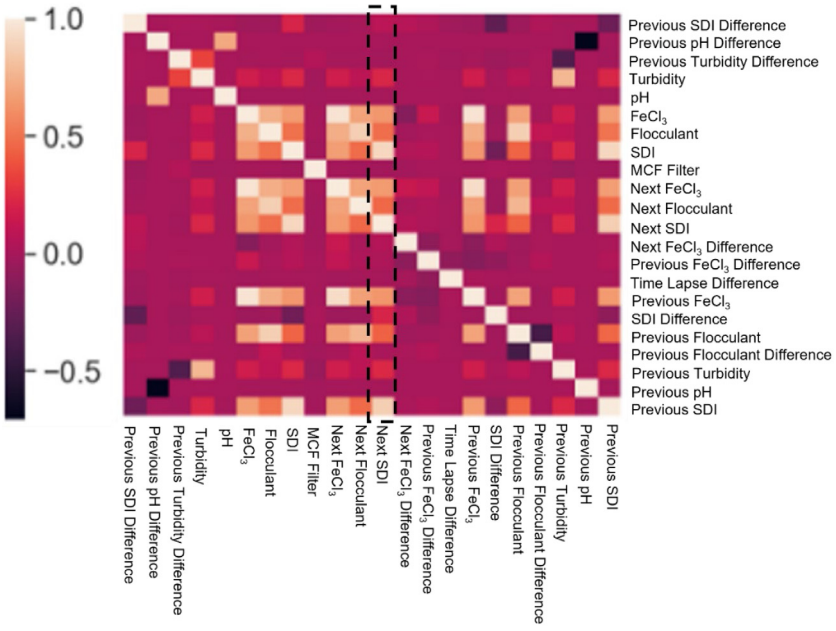


Fig. 2. Pearson correlation of each parameter with the other parameters.

Table 1
Data used for estimating SDI after micron cartridge filter.

Type	Data	Specific
Operational Data	Raw Parameters	Datetime turbidity pH ferric polymer sdi_before_mcf sdi fe_before_mcf fe_after_mcf
Data Processing	Additional Calculated Parameters	next_SDI next_ferric next_polymer diff_ferric_next last_ferric diff_ferric_last last_polymer diff_polymer_last diff_sdi sdi_before diff_sdi_before turbidity_before diff_turbidity_before pH_before diff_pH_before sdi_diff

3.3. Artificial neural network architecture definition

Artificial neural networks are bioinspired algorithms built based on the human brain which contains connected neurons or neural networks (Orr & Müller, 1998). Each layer contains a single or multiple node(s) which is called Neuron. Inside the Neuron, there is a function to perform a mathematical operation. We specifically construct the artificial neural network using a dense layer. The dense layer is the term when the node is connected to all neurons in the preceding layer. In our model development, we utilize the sequential model to construct a layer-by-layer model. Sequential models are linear stacks of layers where one-layer leads to the next. The sequential API is straightforward to implement and only requires the previous layer's output as the input to the next layer (Chollet, 2020), April 12). In other words, Sequential API uses a plain stack of the layer where each layer has one input and output tensor which cannot share layers. We create one input layer followed by four hidden layers. The input layer takes in the data and pre-processes it. The hidden layer performs a nonlinear transformation of input and the output layer takes the results from the hidden layer, transforms them, and gives the final output. To optimize the architecture of our artificial neural network, we adjust several different parameters such as the number of hidden layers, epoch, activation function, optimizer, and batch size.

The input and hidden layers are utilizing a non-linear rectified linear unit (ReLU) as an activation function, where the output is using linear activation function: $O_i = F(w \cdot x_i + b)$, where F represents activation function, w represents the weight data, and b represents a biased value used in machine learning to optimize the model (Nair & Hinton, 2010). To evaluate the model during the training, we employ the mean squared error as the loss function. We further applied the optimization function that optimizes the weights of input based on prediction and the loss function, called adaptive moment estimation (*Adam*). *Adam* is a first-order gradient algorithm based on adaptive estimates of lower-order moments (Kingma Diederik & Adam, 2014; Reddi, Kale, & Kumar, 2019). To evaluate the performance of the model after training, we used mean average error, $MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - x_i|$, and mean square error, $MSE = \left(\frac{\sigma}{\sqrt{n}}\right)^2$, where n is number of sample, y_i and x_i are predicted value and actual value, and σ^2 is variance. The code for the developed model is provided in supplemental information (see Table 2).

3.4. Model comparison

We employ three more models: decision tree regression, random forest regressor, and linear regression in order to further compare the use of the ANNs model and assess its efficacy. The decision tree algorithm is algorithm used based on numbers of criteria. The difference between the random forest regressor and decision tree regression is that random forest regressor utilizes multiple decision tree. During the models' training, the data is treated identically like the ANNs model. The models are evaluated by their coefficient of determination (R^2 score) and mean squared error (MSE). We optimize the decision trees and the random forest regressor by adjusting the maximum depth of the trees. Lastly, we compared the testing scores of trainings and test to avoid overfitting.

Table 2
Summary of artificial neural network architecture.

Type	Data	Specific
Input Layer	Parameters used for Input Layer	diff_pH_before diff_turbidity_before turbidity pH ferric sdi_before_mcf diff_ferric_last last_ferric turbidity_before pH_before
Hidden Layer	Number of hidden layers	4
	Number of units in hidden layer 1	256
	Number of units in hidden layer 2	128
	Number of units in hidden layer 3	64
	Number of units in hidden layer 4	64
	Activation function	Rectified Linear Unit (ReLU)
Output Layer	Dependent Variables	sdi_after_mcf
	Number of units	1
	Activation function	Linear
	Error function	Mean average error and mean square error

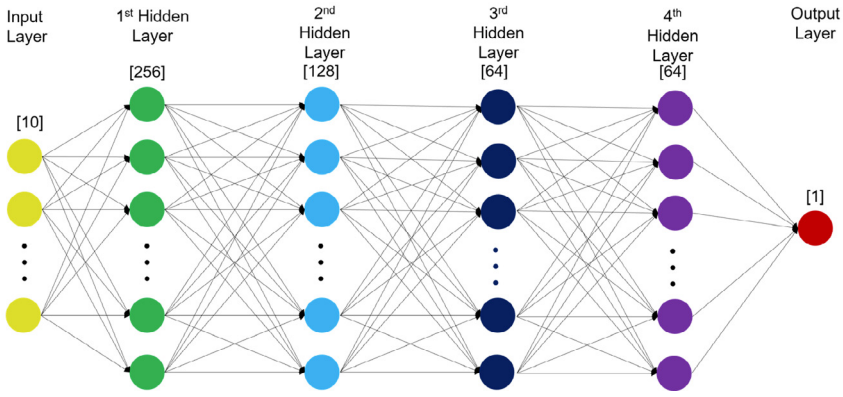


Fig. 3. Schematics of the artificial neural network architecture.

4. Result and discussion

4.1. Artificial neural network forecasting results

There are several factors affecting the quality of ANNs output including its architecture. One main issue with ANNs is overfitting, where the model is memorizing the dataset instead of creating generalization (Domashova et al., 2021). Typically, the higher the epoch, the higher the accuracy. However, if the epoch is too high, the accuracy starts to fall, which indicates overfitting. To address this issue, we conduct a parametric study on the neural-network architecture. We sequentially change epoch, batch size, number of input and hidden layers, and their corresponding nodes. The optimal result is achieved after the training with 500 epochs and batch size of 50. The input layer consists of 10 nodes and 4 hidden layers with 256, 128, 64, and 64 nodes, respectively (Fig. 3).

In the artificial neural network model based on the architectural definition above, we achieved 95.33% accuracy (Fig. 4). This case yields sufficiently minimal mean absolute error (MAE) and mean squared error (MSE), the value of 0.15 and 0.047 respectively. The predicted SDI and actual SDI comparison using this model is deployed in our simulation platform and the corresponding plot is shown below. The high accuracy of testing data indicates that the model can draw the general trend between the input parameters and the output.

To use the algorithm, the user is required to input the feed seawater SDI, pH, turbidity, and previous and proposed coagulant dosing concentration. The user will obtain the predicted value of SDI. Based on this value, the user will decide if they would like to apply the proposed coagulant dosing in the operation or further optimize it.

4.2. Comparative performance with different models

We compare the ANNs model's accuracy with that of three distinct machine learning algorithms: decision tree regression, random forest regressor, and linear regression. The results are displayed in Fig. 5. For decision tree regressor, we found that the optimum maximum depth of the tree is 10. The higher the maximum depth, the better the training score, but lower is test score. It indicates overfitting for maximum depth above 10. With random forest regressor, the maximum depth used is 12. The higher the maximum depth, the higher the training score but the test score is relatively stable. Among the three models, the random forest regressor performs better, with r^2 score of 0.57 compared with 0.49 from linear and decision tree regression. In comparison with these models, the ANNs model is superior in forecasting the SDI value. The comparison of the different training models is presented below (Table 3). Because of its effectiveness, we selected the ANNs model for deployment and further industrialization.

4.3. Model implementation

The developed model is utilized in the SWRO plant to optimize the coagulant dosing. The user inputs the seawater parameters and the theoretical ferric chloride dosing. If the ferric chloride dosing produces the desired SDI value, the dosing rate is applied in the plant. For

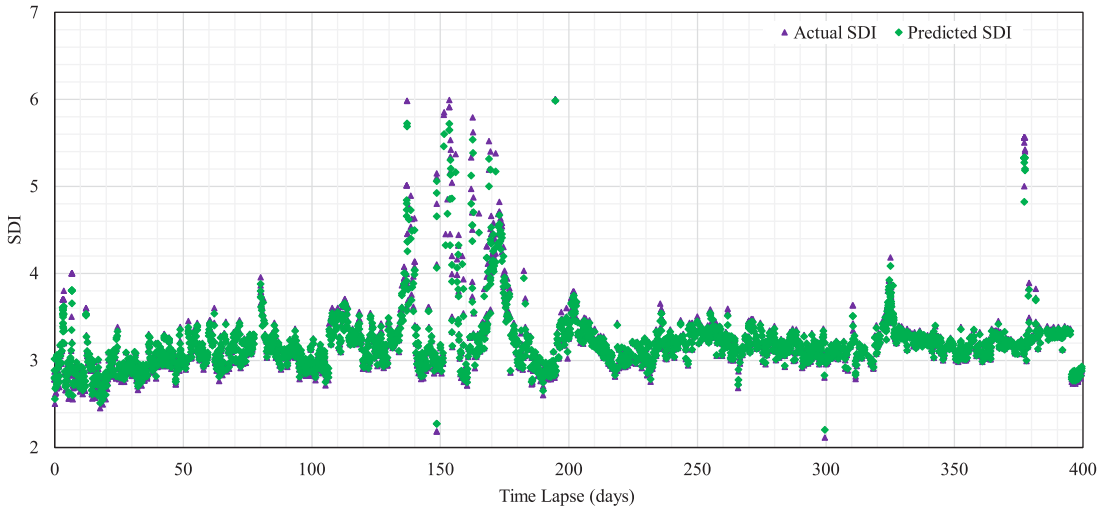


Fig. 4. Comparison of actual SDI and predicted SDI.

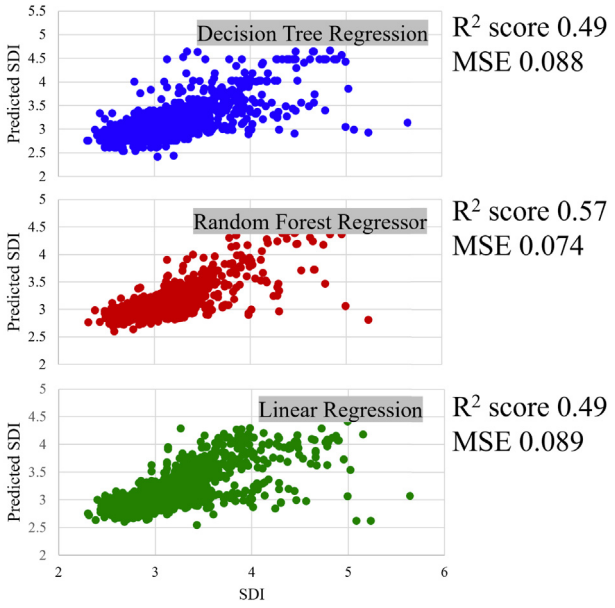


Fig. 5. Comparative performance between decision tree regression, random forest regressor, and linear regression in predicting the SDI of filtrate water.

Table 3
Models' comparison based on the performance score.

Model	R^2 coefficient	Mean Squared Error
Decision Tree Regression	0.49	0.088
Random Forest Regressor	0.57	0.074
Artificial Neural Networks	0.95	0.047
Linear Regression	0.49	0.089

contractual obligations the SDI of the filtered water has to be below 4, if this goes beyond 4, the plant reduces throughput to reduce the rate of filtration to meet this requirement. Fig. 6 shows the SDI of the inlet seawater and filtered water with and without the use of the developed algorithm. Prior to the use of the model a constant dosage rate of ferric chloride was used in the plant and this was increased when the SDI of influent seawater increased. With the use of the algorithm the operators have the flexibility of changing the dosing rate as per the existing seawater condition using the prediction from the model. The total cost of the coagulant depends on its usage and the current market value. While the cost price depends on the supplier contracts,

Table 4
Market value of Ferric Chloride in KSA.

Year	Month	Unit of measurement	Unit price (SAR)
2021	March April May Jun Jul Aug Sep Oct Nov Dec	kg	0.77
2022	Jan Feb		0.84

the usage depends on the plant operators. Table 4 below shows the market value of Ferric Chloride ($FeCl_3$) in KSA. The cost of coagulant dosing is calculated based on the equation $[FeCl_3]_{Total} = \frac{[FeCl_3] \times Q_{FeCl_3}}{10^6}$, where $[FeCl_3]_{Total}$ is total chemical consumption in tonnes/day, $[FeCl_3]$ is coagulant concentration in ppm and Q_{FeCl_3} is coagulant flowrate in m^3/day. The total cost of $FeCl_3$ is defined as $TotalCost_{FeCl_3} = Price_{FeCl_3} \times [FeCl_3]_{Total}$, where $TotalCost_{FeCl_3}$ is the total cost of $FeCl_3$ in SAR/day, $Price_{FeCl_3}$ is coagulant price per kg in SAR/kg.

The total consumption of coagulant for each month, when dosed as per the algorithm is compared with its consumption for the corresponding month in the previous year without the use of the algorithm (Fig. 7). The developed algorithm was deployed to the plant operators. The plant operators adjusted the coagulant dosing rate daily based on the prediction of the algorithm. Upon one year of implementation, the coagulant consumption was reduced by 11.7% and thereby reducing operational expense on chemical consumption by 325,000 SAR. Furthermore, in the typical winter season (Nov-April), the chemical consumption is significantly higher than the necessary value. In contrast, during summer season (May-October), the chemical consumption is near the optimal value. Hence, significant reduction and cost savings is realized during the winter season. The cost saving based on the model implementation offsets the software development, deployment and industrialization costs. In addition, the successful implementation of the ANNs model indicates that the model is unlikely over-fitted.

5. Conclusion

In summary, this study demonstrates the utility of the ANNs algorithm to minimize the coagulant consumption in a large-scale desalination plant by predicting SDI of seawater after the coagulant

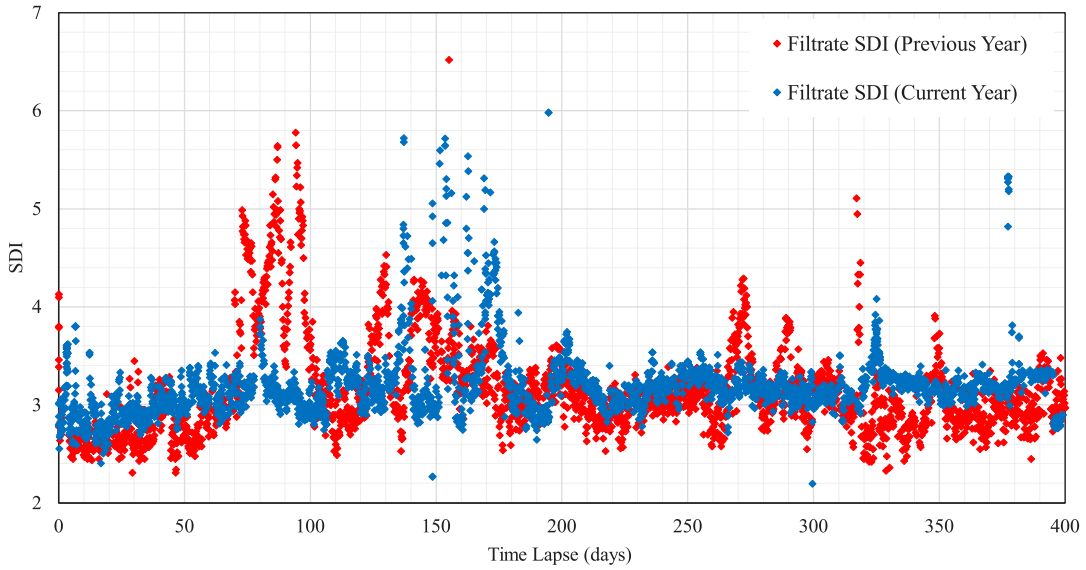


Fig. 6. Comparison of filtrate SDI previous year (without algorithm) and filtrate SDI current year (with algorithm).

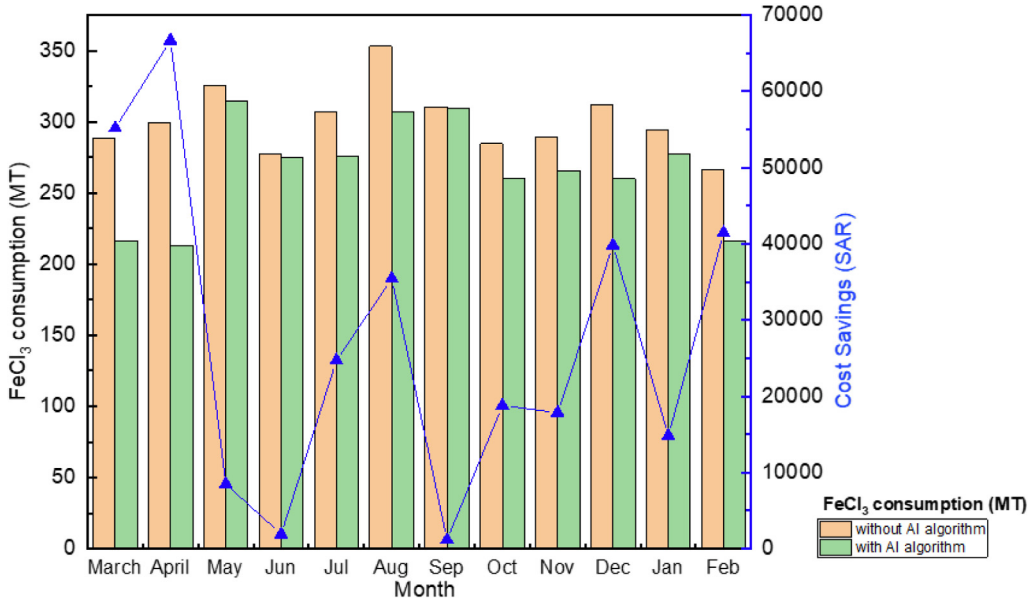


Fig. 7. Comparison of monthly coagulant consumption with and without the use of the AI algorithm and the subsequent cost saving.

dosing. Such a concept has not yet been extensively used at an industrial size desalination plant. Prior to the model development, to remove irrelevant parameters, we employ Pearson's correlation. Upon the determination of the relevant parameter, the parametric study of artificial neural network architecture is done. For predicting the SDI of seawater after coagulant dosing, the ANNs model had an accuracy of 95% compared to the random forest regressor model's accuracy of 57%, which performed the best among the three machine learning techniques used. The ANNs algorithm can accurately predict the SDI value after coagulant dosing, and its subsequent industrialization resulted in reduction of coagulant consumption by 11.7% and thus reduced the operational cost for the desalination plant. The significant chemical and cost-saving heralds the potential of widespread implementation of such ANNs model to optimize other desalination plants and even newer pretreatment technologies such as Ceramic Ultrafiltration. In the future, it would be potentially beneficial to use likewise machine learning and ANNs model to minimize power and other chemicals consumption based on plant performance forecasting.

CRediT authorship contribution statement

Muhammad Ghifari Ridwan: Validation, Formal analysis, Methodology, Writing - review & editing. **Thomas Altmann:** Conceptualization, Writing – review & editing. **Ahmed Yousry:** Methodology, Writing - review & editing. **Ratul Das:** Data curation, Conceptualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.mlwa.2023.100475.

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